

Ethan Randolph Dibble

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Education

Portland State University **2022 – Present**
BS in Computer Science and Mathematics
GPA 3.98/4.0 : President's list 2022 – present *Expected 3/2027*

Technical Skills

Programming: C, C++, Go, Rust, Python, Bash, Lua, Haskell, JavaScript, GLSL, SQL (PostgreSQL, SQLite)
Operating Systems: Linux (Workstation/Server/Pentest), Windows (Workstation)
Developer Tools: Git, GitHub Actions, Make, VSCode, Vim, Agile, Docker/Podman

Personal Projects

[Rayt-rs](#) **Rust** **8/2025 – Present**

- Building a multithreaded software ray tracer that can render images with reflections in less than 5 seconds.
- Extended the system with multiple samplers and materials, as well as TOML deserialization.

[OXID-8](#) **Rust, HTML, CSS, WGS** **4/2025 – 7/2025**

- Engineered and published a fully documented CHIP-8 interpreter core library crate.
- Built desktop and browser renderers (with phone support) using Wgpu, and a TUI renderer using Ratatui.
- Wrote a test suite to validate cpu instruction execution; for example, drawing to a virtual screen.

[Planet](#) **JavaScript, WebGL2, GLSL** **12/2024 – 2/2025**

- Recursively subdivided an octahedron and performed texture mapping with spherical coordinates.
- Wrote artistic light attenuation shaders and loaded them via reusable shader and WebGL2 utilities.

[Technical Blog and Portfolio Website](#) **JavaScript, HTML, CSS** **12/2024 – Present**

- Utilized the JavaScript Fetch API to fill content dynamically through templates.
- Created a custom fork of github-readme-stats that adds a columns parameter to the compact language graph.
- Wrote a Bash script to automate bundling and deployment.

Academic Projects

[Recipe Server](#) **Rust, HTML, CSS** **4/2025 – 6/2025**

- Utilized Axum, Askama, Leptos, and Sqlx to develop a full-stack recipe server with core API functionality.
- Containerized the backend using podman and ran a Leptos frontend using the API.
- Documented the API using the OpenAPI standard to improve usability.

[Music Genre Classification \(Team Project\)](#) **Python & Flask, HTML, CSS** **6/2024 – 8/2024**

- Personally developed a CNN feature extractor to preprocess audio files into compact features for use by the team as input for separate classification models.
- As a leader, I directed bi-weekly team meetings and text communication to track progress and next steps.
- Co-Authored a research paper detailing our finding and the efficacy of our models.
- Deployed a Flask website on AWS, publishing our best classification model with spectrogram images.

MERN Stack Chat App (Team Project): **JavaScript, HTML, CSS** **4/2024 – 6/2024**

- Developed a real-time chat application with a RESTful API and user encryption using Google OAuth2.
- Practiced AGILE methodologies and engaged in weekly standups to track progress and next steps.
- Utilized a microservices architecture and deployed each service through independent Docker containers.